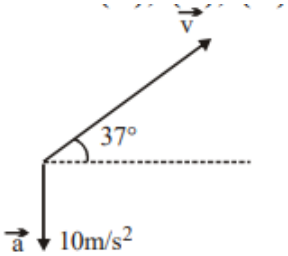
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Just after firing, a bullet is found to move at an angle of 37° to horizontal. Its acceleration is $10\ \mathrm{m/s^2}$ downwards. Find the component of acceleration in the direction of the velocity.



(A) $-6\ \mathrm{m/s^2}$

(B) $-4\ \mathrm{m/s^2}$

(C) $-8\ \mathrm{m/s^2}$

(D) $-5\ \mathrm{m/s^2}$

Q.2

A man rows a boat with a speed of $18\ \mathrm{km/hr}$. in northwest direction. The shoreline makes an angle of 15° south of west. Obtain the component of the velocity of the boat along the shoreline

(A) $9\ \mathrm{km/hr}$

(B) $18\frac{\sqrt{3}}{2}\ \mathrm{km/hr}$

(C) $18\cos 15^\circ \mathrm{km/hr}$

(D) $18\cos 75^\circ \mathrm{km/hr}$

Q.3

A dust particle oscillates in air with a time period which depends on atmospheric pressure P, density of air d and energy of the particle E, then time period is proportional to

(A) $P^{-\frac{5}{6}}d^{\frac{1}{2}}E^{\frac{1}{3}}$

(B) $P^{\frac{1}{2}}d^{3}E^{-2}$

(C) $P^{-\frac{1}{3}}d^{\frac{1}{2}}E^{2}$

(D) $P^{-2}d^{-\frac{1}{2}}E^{-3}$

Q.4

Two resistances of 400Ω and 800Ω connected in series with a 6 volt battery of negligible internal resistance. A voltmeter of resistance $10,000\Omega$ is used to measure the potential difference across 400Ω. The error in the measurement of potential difference in volts approximately is

(A) 0.01

(B) 0.02

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(C) 0.04

(D) 0.05

Q.5

A liquid drop placed on a horizontal plane has a near spherical shape (slightly flattened due to gravity). Let R be the radius of its largest horizontal section. A small disturbance causes the drop to vibrate with frequency v about its equilibrium shape. By dimensional analysis the ratio $\frac{v}{\sqrt{\frac{\sigma}{\rho R^3}}}$ can be (Here σ is surface tension; ρ is density, g is acceleration due to gravity, and k is an arbitrary dimensionless constant)

(A) $\frac{k\rho g R^2}{\sigma}$

(B) $\frac{k\rho R^3}{g\sigma}$

(C) $\frac{k\rho R^2}{g\sigma}$

(D) $\frac{k\rho}{g\sigma}$

Q.6

In a system of units if force (F), acceleration (A) and time (T) are taken as fundamental units then the dimensional formula of energy is -

(A) $\mathrm{FA}^2\ \mathrm{T}$

(B) FAT^2

(C) $\mathrm{F}^2\mathrm{AT}$

(D) FAT

Q.7

The pitch of a screw gauge is $1\ \mathrm{mm}$ and there are 100 division on its circular scale. When nothing is put in between its jaws, the zero of the circular scale lies 4 divisions below the reference line. When a steel wire is placed between the jaws, two main scale divisions are clearly visible and 67 divisions on the circular scale are observed. The diameter of the wire is

(A) $2.71\ \mathrm{mm}$

(B) $2.67\ \mathrm{mm}$

(C) $2.63\ \mathrm{mm}$

(D) $2.65\ \mathrm{mm}$

Q.8

For a body moving along x-axis, the distance travelled by body from a reference point is given as function of time t as $x = at^2 + b$, where a and b are constants, then the dimension of $\sqrt{ab}$ is same as

(A) Speed

(B) Distance travelled

(C) Acceleration

(D) None of these

Q.9

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If energy (E), velocity (v) and force (F) be taken as fundamental quantity, then which of the following does not match the dimensions of mass -

(A) Ev^2

(B) Ev^{-2}

(C) Fv^{-1}

(D) Fv^{-2}

Q.10

A physical quantity x depends on quantities y and z as follows: $x = Ay + B\tan Cz$, where A, B and C are constants. Which of the following have the same dimensions

(A) x and B

(B) C and z^{-1}

(C) y and $\mathrm{B/A}$

(D) x and A

Q.11

Two forces $\overrightarrow{\mathrm{A}}$ and $\overrightarrow{\mathrm{B}}$ have resultant $\overrightarrow{\mathrm{R}}_1$. If $\overrightarrow{\mathrm{B}}$ is doubled the new resultant $\overrightarrow{\mathrm{R}}_2$ is at right angles to $\overrightarrow{\mathrm{A}}$. Choose the correct options -

(A) $\mathrm{R}_1 = \mathrm{B}$

(B) $\mathrm{R}_2 = \sqrt{4\,\mathrm{B}^2 - \mathrm{A}^2}$

(C) $\mathrm{R}_1 = 2\,\mathrm{B}$

(D) $\mathrm{R}_2 = \sqrt{2\,\mathrm{B}^2 - \mathrm{A}^2}$

Q.12

Passage:
A ball starting from origin moves for 1 second at constant velocity $(\hat{i} + 2\hat{j} + 2\hat{k})\mathrm{m/s}$.
If then bounces of a wall, and moves for 2 second at new constant velocity $(\hat{i} + 2\hat{j} - 2\hat{k})\mathrm{m/s}$.
Question: What is the change in velocity due to collision with wall -

(A) zero

(B) $4\hat{\mathrm{k}}\ \mathrm{m/s}$

(C) $-4\hat{\mathrm{k}}\ \mathrm{m/s}$

(D) $2\hat{\mathrm{i}} + 4\hat{\mathrm{j}}\ \mathrm{m/s}$

Q.13

Passage:
A ball starting from origin moves for 1 second at constant velocity $(\hat{i} + 2\hat{j} + 2\hat{k})\mathrm{m/s}$.
If then bounces of a wall, and moves for 2 second at new constant velocity $(\hat{i} + 2\hat{j} - 2\hat{k})\mathrm{m/s}$.
Question: At the end of motion described above the displacement of ball from the origin is -

(A) zero

(B) $3\ \mathrm{m}$

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(C) $6\ \mathrm{m}$

(D) $7\ \mathrm{m}$

Q.14

Passage:

A ball starting from origin moves for 1 second at constant velocity $(\hat{i}+2\hat{j}+2\hat{k})\mathrm{m/s}$.

If then bounces of a wall, and moves for 2 second at new constant velocity $(\hat{i}+2\hat{j}-2\hat{k})\mathrm{m/s}$.

Question: What is the average velocity for the period of above described 3 seconds ?

(A) $\hat{i}-2\hat{j}+\frac{2\hat{k}}{3}(\ \mathrm{m/s})$

(B) $\hat{i}+2\hat{j}-\frac{2\hat{k}}{3}(m/s)$

(C) $\hat{\mathrm{i}}+2\hat{\mathrm{j}}-6\hat{\mathrm{k}}(\mathrm{m/s})$

(D) $\hat{i}+2\hat{j}-4\hat{k}(\ \mathrm{m/s})$

Q.15

Passage:

The limit of a derived quantity in terms of necessary basic units is called dimensional formula and the raised powers on the basic units are dimensions.

The dimension of physical quantity on the left hand side of dimensional equation should equal to the net dimensions of all physical quantities on the right hand side of it. Only those physical quantities can be added or subtracted which have same dimensions.

Question: The equation of a wave is given by $Y=A\sin\omega\left(\frac{x}{v}-k\right)$, where ω is the angular velocity and v is the linear velocity. The dimension of k is -

(A) LT

(B) T

(C) T^{-1}

(D) T^{2}

Q.16

Passage:

The limit of a derived quantity in terms of necessary basic units is called dimensional formula and the raised powers on the basic units are dimensions.

The dimension of physical quantity on the left hand side of dimensional equation should equal to the net dimensions of all physical quantities on the right hand side of it. Only those physical quantities can be added or subtracted which have same dimensions.

Question: The time period of a body under SHM i.e. presented by $P^aD^bS^c$; where P is pressure, D is density and S is surface tension. The value of a,b and c are $-$

(A) $-\frac{3}{2},\frac{1}{2},1$

(B) $-1,-2,3$

(C) $\frac{1}{2},-\frac{3}{2},-\frac{1}{2}$

(D) $1,2,\frac{1}{3}$

Q.17

Passage:

The limit of a derived quantity in terms of necessary basic units is called dimensional formula and the raised powers on the basic

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units are dimensions.

The dimension of physical quantity on the left hand side of dimensional equation should equal to the net dimensions of all physical quantities on the right hand side of it. Only those physical quantities can be added or subtracted which have same dimensions.

Question: The velocity of a freely falling body changes as $g^p h^r$ where g is acceleration due to gravity and h is the height. The values of p and r are -

(A) $1, \frac{1}{2}$

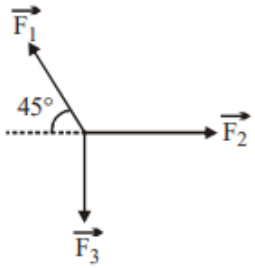
(B) $\frac{1}{2}, \frac{1}{2}$

(C) $\frac{1}{2}, 1$

(D) 1,2

Q.18

Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ are represented as shown. Each of them is of equal magnitude.



Column I (Combination)	Column II (Approximate Direction)
(A) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3$	(p) ↖
(B) $\vec{F}_1 - \vec{F}_2 + \vec{F}_3$	(q) ↘
(C) $\vec{F}_1 - \vec{F}_2 - \vec{F}_3$	(r) ↙
(D) $\vec{F}_2 - \vec{F}_1 - \vec{F}_3$	(s) ↗

(A) $A-(q)B-(r)C-(p)D-(s)$

(B) $A-(q)B-(r)C-(s)D-(p)$

(C) $A-(p)B-(s)C-(q)D-(r)$

(D) $A-(s)B-(p)C-(q)D-(r)$

Q.19

If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors such that $\vec{a}+\vec{b}+\vec{c}=0$, then $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$ is equal to ____ (Answer the absolute value)

Q.20

If P represents radiation pressure, c represents speed of light and Q represents radiation energy striking a unit area per second, then non-zero integers x, y and z such that $P^x Q^y c^z$ is dimensionless, are $x=1, y=-a, z=b$. Find the value of $a+b$.

Answers & Solutions

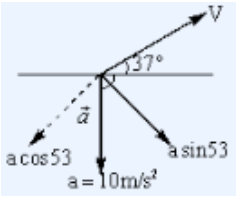
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Q.1 Answer:

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Solution:

Component of acceleration in the direction of velocity $= -a\cos 53 = -10 \times \frac{3}{5} = -6\ \mathrm{m/s^2}$



Q.2 Answer:

$9\ \mathrm{km/hr}$

Solution:

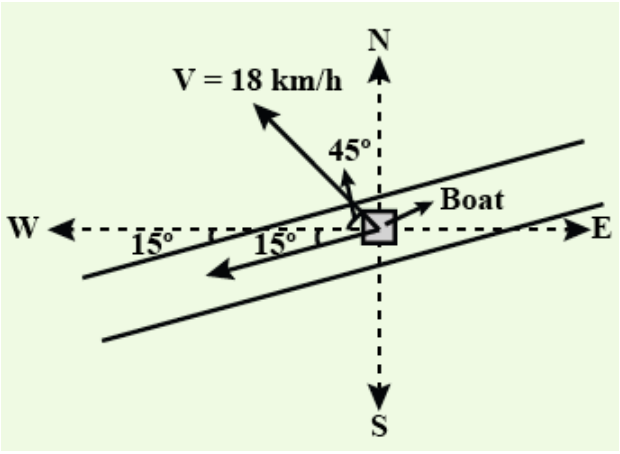
The speed of boat $= 18\ \mathrm{km/hr}$ **in north west direction. That is velocity makes an angle of** 45° **with west.**

We have to find the velocity along shoreline or parallel to river.

We can see in figure that the velocity V makes an angle $(45 + 15) = 60^\circ$ **with the river.**

So,

$$\mathrm{V}\cos\theta = \mathrm{V}\cos 60^\circ$$
$$= 18 \times \frac{1}{2}$$
$$= 9\ \mathrm{km/hr}$$



Q.3 Answer:

$P^{-\frac{5}{6}} d^{\frac{1}{2}} E^{\frac{1}{3}}$

Solution:

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$T = kp^x d^y E^z$

$[T] = \left[ML^{-1}T^{-2}\right]^x \left[ML^{-3}\right]^y \left[ML^2T^{-2}\right]^z$

$x + y + z = 0 \quad \ldots(1)$

$-x - 3y + 2z = 0 \quad \ldots(2)$

$-2x - 2z = 1 \quad \ldots(3)$

$x + z = -\frac{1}{2} \quad \ldots(4)$

$-y = -\frac{1}{2}$

$\Rightarrow \quad y = \frac{1}{2}$

by equation (2) $-x - \frac{3}{2} + 2z = 0$

$-x + 2z = \frac{3}{2}$ **...(5)**

Adding (4) & (5)

$3z = 1, z = \frac{1}{3}$

$\Rightarrow \quad x = -\frac{1}{2} - \frac{1}{3} = -\frac{5}{6}$

∴ (a)

Q.4 Answer:

0.04

Solution:

$I = \frac{6}{400+800} = 5 \times 10^{-3}$ A

∴ **Voltage drop across** $400\Omega = 5 \times 10^{-3} \times 400 = 2$ V

Because of the presence of the voltmeter having resistance $G = 10,000\Omega$ **in parallel with** 400Ω**, the effective resistance is**

$\frac{400 \times 10,000}{10,400} = \frac{10,000}{26}\Omega$

∴ **Voltage measured** $= \frac{10,000}{26} \times 5 \times 10^{-3} = \frac{50}{26}$ V

∴ **Relative error in the measurement** $= \frac{1}{26} = 0.04$ **volt**

∴ **(c)**

Q.5 Answer:

$\frac{k\rho g R^2}{\sigma}$

Solution:

$\frac{v}{\sqrt{\sigma/\rho R^3}} = v\sqrt{\frac{\rho R^3}{\sigma}} = T^{-1}\left[\frac{ML^{-3}L^3}{MT^{-2}}\right]^{1/2} = T^{-1}T = 1$

$\frac{v}{\sqrt{\sigma/\rho R^3}} = K \Rightarrow v^2\frac{\rho R^3}{\sigma} = K^2$

$\Rightarrow K^2 = \frac{\rho R^3}{\sigma}T^{-2} = \frac{\rho R^2}{\sigma}RT^{-2} \Rightarrow K = \frac{\rho R^2}{\sigma}g.$

Q.6 Answer:

FAT ^{2}

Solution:

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$E = KF^aA^bT^c$

$[ML^2T^{-2}] = [MLT^{-2}]^a [LT^{-2}]^b [T]^c$

$[ML^2T^{-2}] = [M^aL^{a+b}T^{-2a-2b+c}]$

$\therefore \quad a = 1, a + b = 2 \Rightarrow b = 1$

and $-2a - 2b + c = -2 \Rightarrow c = 2 \quad \therefore E = KFAT^2$

Q.7 Answer:

2.63 mm

Solution:

$p = 1$ mm, $N = 100$

Least count,

$C = \frac{P}{N} = \frac{1 \text{ mm}}{100} = 0.01$ mm

The instrument has a positive zero error $e = +NC = +4 \times 0.01 = +0.04$ mm

Main scale reading is $2 \times (1 \text{ mm}) = 2$ mm

Circular scale reading is $67(0.01) = 0.67$ mm

$\therefore$ **observed reading is** $R_0 = 2 + 0.67 = 2.67$ mm

So true reading $= R_0 - e = 2.63$ mm

$\therefore$ **(c)**

Q.8 Answer:

Speed

Solution:

As $x = at^2 + b$ **therefore** $[a] = [LT^{-2}]$ **and** $[b] = [L] \quad \therefore [ab] = [L^2T^{-2}] = [v^2]$

$\therefore \sqrt{(ab)} = v$

Q.9 Answer:

Ev^2

Solution:

Let $m \propto E^x v^y F^z$

By substituting the following dimensions:

$[E] = [ML^2T^{-2}], [v] = [LT^{-1}], [F] = [MLT^{-2}]$

and by equating the both sides $x = 1, y = -2, z = 0$. **So** $[m] = [Ev^{-2}]$

Q.10 Answer:

x and B

Solution:

$x = Ay + B \tan Cz$

From the dimensional homogeneity

$[x] = [Ay] = [B] \Rightarrow \left[\frac{x}{A}\right] = [y] = \left[\frac{B}{A}\right]$

$[Cz] = [M^0L^0T^0] =$ Dimension less

x and B; C and Z^{-1}**; y and** $\frac{B}{A}$ **have the same dimension but x and A have the different dimensions.**

Q.11 Answer:

$R_1 = B$

Solution:

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Let angle between $\vec{A}$ and $\vec{B}$ be θ, so angle between $\overrightarrow{\mathrm{A}}$ and $2\overrightarrow{\mathrm{B}}$ is also θ.

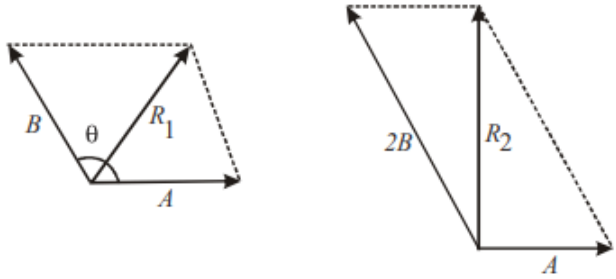
$\Rightarrow \quad R_1^2 = A^2 + B^2 + 2AB\cos\theta \ldots (1)$

$$\text{and } \tan 90^\circ = \frac{2B\sin\theta}{A + 2B\cos\theta} \to \infty$$

$\Rightarrow \quad \mathrm{A} + 2\,\mathrm{B}\cos\theta = 0 \Rightarrow \cos\theta = -\frac{\mathrm{A}}{2\,\mathrm{B}} \ldots\ldots$ (2)

Substituting (2) in (1), we get, $R_1 = B$

Also, $R_2^2 + A^2 = (2B)^2 = 4B^2 \Rightarrow R_2 = \sqrt{4B^2 - A^2}$



Q.12 Answer:

$-4\hat{\mathrm{k}}\ \mathrm{m/s}$

Solution:

Change in velocity $= (\hat{\mathrm{i}} + 2\hat{\mathrm{j}} - 2\hat{\mathrm{k}}) - (\hat{\mathrm{i}} + 2\hat{\mathrm{j}} + 2\hat{\mathrm{k}}) = -4\hat{\mathrm{k}}\ \mathrm{m/s}$

Q.13 Answer:

$7\ \mathrm{m}$

Solution:

At central bright fringe the waves from slits are in phase and following bright fringes having a difference of 2π.

$4\pi = 2 \times 2\pi \Rightarrow$ 2nd order bright fringe that is 4.

Q.14 Answer:

$\hat{i} + 2\hat{j} - \frac{2\hat{k}}{3}(m/s)$

Solution:

Change in velocity $= (\hat{\mathrm{i}} + 2\hat{\mathrm{j}} - 2\hat{\mathrm{k}}) - (\hat{\mathrm{i}} + 2\hat{\mathrm{j}} + 2\hat{\mathrm{k}}) = -4\hat{\mathrm{k}}$

Final position vector $= (\hat{i} + 2\hat{j} + 2\hat{k}) \times 1 + (\hat{i} + 2\hat{j} - 2\hat{k}) \times 2 = 3\hat{i} + 6\hat{j} - 2\hat{k}$

Displacement $= \sqrt{3^2 + 6^2 + 2^2} = 7\ \mathrm{m}$

Average velocity $= \frac{\text{Total displacement}}{\text{Total time}} = \frac{3\hat{\mathrm{i}} + 6\hat{\mathrm{j}} - 2\hat{\mathrm{k}}}{3} = \left(\hat{\mathrm{i}} + 2\hat{\mathrm{j}} - \frac{2}{3}\hat{\mathrm{k}}\right)\ \mathrm{m/s}$

Q.15 Answer:

T

Solution:

From the principle of homogenity $\left(\frac{\mathrm{x}}{\mathrm{v}}\right)$ has dimensions of T.

$\therefore$ Dimension of k is T.

Q.16 Answer:

$-\frac{3}{2}, \frac{1}{2}, 1$

Solution:

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By substituting the dimension of each quantity we get $\mathrm{T}=\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-2}\right]^{\mathrm{a}}\left[\mathrm{L}^{-3}\mathrm{M}\right]^{\mathrm{b}}\left[\mathrm{MT}^{-2}\right]^{\mathrm{c}}$

By solving we get $a=-3/2, b=1/2$ **and** $c=1$

Q.17 Answer:

$\frac{1}{2}, \frac{1}{2}$

Solution:

$\mathrm{v} \propto \mathrm{g}^{\mathrm{p}} \mathrm{h}^{\mathrm{q}}$ (given)

By substituting the dimension of each quantity and comparing the powers in both sides

we get $\left[\mathrm{LT}^{-1}\right]=\left[\mathrm{LT}^{-2}\right]^{\mathrm{p}}[\mathrm{L}]^{\mathrm{q}}$

$\Rightarrow \mathrm{p}+\mathrm{q}=1,-2\mathrm{p}=-1$

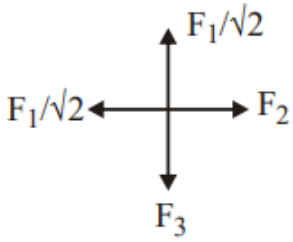
$\therefore p=\frac{1}{2}, q=\frac{1}{2}$

Q.18 Answer:

$A-(q)B-(r)C-(p)D-(s)$

Solution:

$A-(q)B-(r)C-(p)D-(s)$



$\overrightarrow{\mathrm{F}}_1+\overrightarrow{\mathrm{F}}_2+\overrightarrow{\mathrm{F}}_3=\left(-\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\hat{\mathrm{i}}+\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\hat{\mathrm{j}}\right)+\overrightarrow{\mathrm{F}}_2\hat{\mathrm{i}}-\overrightarrow{\mathrm{F}}_3\hat{\mathrm{j}}=\left(\overrightarrow{\mathrm{F}}_2-\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\right)\hat{\mathrm{i}}-\left(\overrightarrow{\mathrm{F}}_3-\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\right)\hat{\mathrm{j}} \Rightarrow$ **In 4th quadrant**

$\overrightarrow{\mathrm{F}}_1-\overrightarrow{\mathrm{F}}_2+\overrightarrow{\mathrm{F}}_3=\left(-\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\hat{\mathrm{i}}+\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\hat{\mathrm{j}}\right)-\overrightarrow{\mathrm{F}}_2\hat{\mathrm{i}}-\overrightarrow{\mathrm{F}}_3\hat{\mathrm{j}}=-\left(\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}+\overrightarrow{\mathrm{F}}_2}\right)\hat{\mathrm{i}}-\left(\overrightarrow{\mathrm{F}}_3-\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\right)\hat{\mathrm{j}} \Rightarrow$ **In** 3rd **quadrant**

$\overrightarrow{\mathrm{F}}_1-\overrightarrow{\mathrm{F}}_2-\overrightarrow{\mathrm{F}}_3=\left(-\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\hat{\mathrm{i}}+\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\hat{\mathrm{j}}\right)-\overrightarrow{\mathrm{F}}_2\hat{\mathrm{i}}+\overrightarrow{\mathrm{F}}_3\hat{\mathrm{j}}=-\left(\overrightarrow{\mathrm{F}}_2+\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\right)\hat{\mathrm{i}}+\left(\overrightarrow{\mathrm{F}}_3+\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\right)\hat{\mathrm{j}} \Rightarrow$ **In** 2nd **quadrant**

$\overrightarrow{\mathrm{F}}_2-\overrightarrow{\mathrm{F}}_1-\overrightarrow{\mathrm{F}}_3=\overrightarrow{\mathrm{F}}_2\hat{\mathrm{i}}-\left(-\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\hat{\mathrm{i}}+\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\hat{\mathrm{j}}\right)+\overrightarrow{\mathrm{F}}_3\hat{\mathrm{j}}=\left(\overrightarrow{\mathrm{F}}_2+\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\right)\hat{\mathrm{i}}+\left(\overrightarrow{\mathrm{F}}_3-\frac{\overrightarrow{\mathrm{F}}_1}{\sqrt{2}}\right)\hat{\mathrm{j}} \Rightarrow$ **In** 1st **quadrant**

Q.19 Answer:

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Solution:

$\vec{a}+\vec{b}+\vec{c}=0$

$(\vec{a}+\vec{b}+\vec{c})\cdot(\vec{a}+\vec{b}+\vec{c})=0\cdot(\vec{a}+\vec{b}+\vec{c})$

$a^2+b^2+c^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=0$

$\therefore \vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}=-3/2$

So answer is 1.5.

Q.20 Answer:

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Solution:

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By substituting the dimension of given quantities

$$\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-2}\right]^{\mathrm{x}}\left[\mathrm{MT}^{-3}\right]^{\mathrm{y}}\left[\mathrm{LT}^{-1}\right]^{\mathrm{z}}=[\mathrm{MLT}]^{0}$$

By comparing the power of $\mathrm{M}, \mathrm{L}, \mathrm{T}$ **in both sides**

$$x+y=0 \ldots(i)$$

$$-x+z=0 \ldots(ii)$$

$$-2x-3y-z=0 \ldots(iii)$$

satisfying (i), (ii) and (iii) corresponds to $x=1, y=-1, z=1$

The only values of x, y, z **satisfying (i), (ii) and (iii) corresponds to** $x=1, y=-1, z=1$

$\therefore a=1$ **as well as** $b=1$ **which implies** $a+b=2$

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